Introduction to Mathematics for Deep Learning: Midterm Solutions

1. **Mix of Questions** *(50 points, 10 points each)*

   1) Here is a statement about pairwise independence and joint independence. Let X, Y_1, and Y_2 be binary random variables. If $I(X;Y_1) = 0$ and $I(X;Y_2) = 0$, does it follow that $I(X;Y_1,Y_2) = 0$? Prove or provide a counterexample.
   2) Suppose a (non-stationary) Markov chain starts in one of n states, necks down to $k < n$ states, and then fans back to $m > k$ states. Thus $X_1 - X_2 - X_3$ forms a Markov chain, where $X_1 \in \{1, 2, \ldots, n\}$, $X_2 \in \{1, 2, \ldots, k\}$, $X_3 \in \{1, 2, \ldots, m\}$, and $p(x_1, x_2, x_3) = p(x_1)p(x_2|x_1)p(x_3|x_2)$. Show that the dependence of X_1 and X_3 is limited by the bottleneck by proving that $I(X_1; X_3) \leq \log k$.
   3) For independent continuous random variables X and Y,
   show that $h(X + Y) \geq h(X)$. [hint: Consider $h(X + Y|Y)$]
   4) The *World Series* is a seven-game series that terminates as soon as either team wins four games. Let X be the random variable representing the full outcome of the series between teams A and B; for example, possible values of X include sequences such as AAAA, BABABAB, and BBBAAAA. Assuming that: i) the two teams are equally matched (each game is won by A or B with probability 1/2), and ii) all games are independent. Compute $H(X)$.
   5) Let $\{X_t\}_{t\geq 0}$ be a 1-st order Markov process on the finite state space $\{0, 1, \ldots, K\}$ with transition probabilities

$$P(X_{t+1} = j|X_t = i) = \begin{cases} 1-\varepsilon & \text{if } i = j \geq 1 \\ \varepsilon & \text{if } j = 0 \text{ and } i \geq 1 \\ 1 & \text{if } i = j = 0 \\ 0 & \text{otherwise} \end{cases}$$

   Assume $0 < \varepsilon < 1$. Compute the *limit distribution* of the process, i.e., π_∞.

Solution:

1) The answer is "no". Although at first the conjecture seems reasonable enough– after all, if Y_1 gives you no information about X, and if Y_2 gives you no information about X, then why should the two of them together give any information? But remember, it is NOT the case that $I(X;Y_1,Y_2) = I(X;Y_1) + I(X;Y_2)$. The chain rule for information says instead that $I(X;Y_1,Y_2) = I(X;Y_1) + I(X;Y_2|Y_1)$. The chain rule gives us reason to be skeptical about the conjecture.

   This problem is reminiscent of the well-known fact in probability that pair-wise independence of three random variables is not sufficient to guarantee that all three

are mutually independent. $I(X;Y_1) = 0$ is equivalent to saying that X and Y_1 are independent. Similarly for X and Y_2. But just because X is pairwise independent with each of Y_1 and Y_2, it does not follow that X is independent of the vector (Y_1, Y_2).

Here is a simple counterexample. Let Y_1 and Y_2 be independent fair coin flips. And let $X = Y_1$ XOR Y_2. X is pairwise independent of both Y_1 and Y_2, but obviously not independent of the vector (Y_1, Y_2), since X is uniquely determined once you know (Y_1, Y_2).

2) From the data processing inequality, and the fact that entropy is maximum for a uniform distribution, we get

$$\begin{aligned} I(X_1; X_3) \leq & I(X_1; X_2) \\ = & H(X_2) - H(X_2 \mid X_1) \\ \leq & H(X_2) \\ \leq & \log k. \end{aligned}$$

3) Since $I(X+Y;Y) \geq 0$, we have

$$\begin{aligned} h(X+Y) \geq & h(X+Y|Y) \\ = & h(X|Y) \\ = & h(X). \end{aligned}$$

where the last equality is due to independence between X and Y.

4) Let Y denote the number of games played, so $Y \in \{4,5,6,7\}$. The series lasts y games if the winning team wins the last game and has exactly 3 wins in the first $y-1$ games. Thus

$$P(Y=y) = 2 \times \frac{\binom{y-1}{3}}{2^y}, \qquad y = 4,5,6,7,$$

where the factor 2 accounts for symmetry between A and B.

y	4	5	6	7
$P(Y=y)$	1/8	1/4	5/16	5/16

Each possible outcome sequence x has probability $P(x) = (1/2)^{|x|}$, and the number of valid sequences of length y is $N_y = 2\binom{y-1}{3}$. Hence

$$H(X) = -\sum_x P(x) \log_2 P(x) = \sum_{y=4}^{7} N_y (1/2)^y \, y = \sum_{y=4}^{7} y \, P(Y=y) = \frac{93}{16} \text{ bits.}$$

5) State 0 is absorbing, since $\mathbb{P}(X_{t+1} = 0 \mid X_t = 0) = 1$. For each $i \in \{1, \ldots, K\}$ the chain either stays at i with probability $1-\varepsilon$ or jumps to 0 with probability ε. There are no transitions among $\{1, \ldots, K\}$ other than self-loops.

Fix $i \in \{1, \ldots, K\}$. Starting from $X_0 = i$, the time to jump to 0 is geometric with success probability ε, hence the probability of *not* having jumped to 0 by time t is $(1-\varepsilon)^t$. Therefore,

$$\mathbb{P}_i(X_t = i) = (1-\varepsilon)^t, \qquad \mathbb{P}_i(X_t = 0) = 1 - (1-\varepsilon)^t,$$

and $\mathbb{P}_i(X_t = j) = 0$ for all $j \notin \{0, i\}$. In particular, $\mathbb{P}_i(X_t = 0) \to 1$ as $t \to \infty$.
Let the initial distribution be $\mu = (\mu_0, \mu_1, \ldots, \mu_K)$, where $\mu_i = \mathbb{P}(X_0 = i)$. By the law of total probability and the decomposition above,

$$\mathbb{P}(X_t = 0) = \mu_0 \cdot 1 \, + \, \sum_{i=1}^{K} \mu_i \big[1 - (1-\varepsilon)^t\big] = 1 - (1-\varepsilon)^t \sum_{i=1}^{K} \mu_i = 1 - (1-\varepsilon)^t (1 - \mu_0).$$

Thus $\mathbb{P}(X_t = 0) \to 1$ as $t \to \infty$. For each $i \geq 1$,

$$\mathbb{P}(X_t = i) = \mu_i (1-\varepsilon)^t \;\longrightarrow\; 0.$$

Therefore, regardless of μ, the distribution of X_t converges to the point mass at 0:

$$\boxed{\lim_{t\to\infty} \mathbb{P}(X_t = 0) = 1, \quad \lim_{t\to\infty} \mathbb{P}(X_t = i) = 0 \;\; (i = 1, \ldots, K).}$$

Equivalently, the unique stationary (and limiting) distribution for $\varepsilon > 0$ is

$$\pi = (1, 0, 0, \ldots, 0).$$

2. **MLE** *(25 points)*

Let $X_1, \ldots, X_n$ be i.i.d. samples from an exponential distribution with parameter $\lambda > 0$, i.e.,

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

(a) Derive the maximum likelihood estimator (MLE) $\widehat{\lambda}_{\text{MLE}}$ of λ. *(10 points)*

(b) Show that $\mathbb{E}[1/\widehat{\lambda}_{\text{MLE}}] = 1/\lambda$. *(5 points)*

(c) Show that $\text{Bias}(\widehat{\lambda}_{\text{MLE}}) > 0$. *(10 points)*

Solution:

(a) The log-likelihood based on $x_1, \ldots, x_n$ is

$$\ell(\lambda) = \sum_{i=1}^{n} \log(\lambda e^{-\lambda x_i}) = n \log \lambda - \lambda \sum_{i=1}^{n} x_i.$$

Differentiating and setting to zero:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0 \quad \Longrightarrow \quad \widehat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\overline{X}},$$

where $\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$.

(b) It is clear that

$$\mathbb{E}[1/\widehat{\lambda}_{\text{MLE}}] = \mathbb{E}[\frac{1}{n}\sum_{i=1}^{n} X_i] = \mathbb{E}[X_i] = 1/\lambda.$$

(c) Since $f(x) = 1/x$ is a convex function on $x > 0$, from Jensen's inequality, we have

$$1/\lambda = \mathbb{E}[1/\widehat{\lambda}_{\text{MLE}}] \geq 1/\mathbb{E}[\widehat{\lambda}_{\text{MLE}}].$$

Note that for strictly convex function $f''(x) > 0$, the equality condition of Jensen's inequality holds only for deterministic random variable. Thus,

$$\text{Bias}(\widehat{\lambda}_{\text{MLE}}) = \mathbb{E}[\widehat{\lambda}_{\text{MLE}}] - \lambda > 0.$$

3. **JS-Divergence** *(25 points)*

Let P and Q be discrete probabilities on the alphabet $\{1, 2, \cdots, K\}$. Relative entropy $D(P||Q)$, can be viewed as a distance between probability distributions. In general it is not symmetric, i.e., there exist P and Q for which $D(P||Q) \neq D(Q||P)$. We can construct symmetric distance measures between two distributions based on relative entropy; one such measure is called the Jensen-Shannon Divergence. For $R = (P + Q)/2$, i.e., R is a new probability distribution on the alphabet $\{1, 2, \cdots, K\}$ such that

$$R(i) = \frac{P(i) + Q(i)}{2} \quad \text{for all } i \in \{1, \ldots, K\}$$

the JS-Divergence is defined as

$$JS(P||Q) = \frac{1}{2}D\left(P\|R\right) + \frac{1}{2}D\left(Q\|R\right).$$

It is easy to check that $JS(P||Q) = JS(Q||P)$.

(a) Let Z be a Bernoulli($\frac{1}{2}$) random variable. Let X be a random variable such that $X|\{Z = 1\}$ has a distribution P while $X|\{Z = 0\}$ has a distribution Q. Find the pmf p_X of random variable X as a function of P, Q, R. *(5 points)*

(b) For the same X and Z in the previous problem, express $JS(P\|Q)$ in terms of $I(X; Z)$. *(10 points)*

(c) What is the maximum value of $JS(P||Q)$ among all possible distributions P and Q. [Hint: Use the previous results.] *(10 points)*

Solution:

(a) We have

$$\begin{aligned} p_X(i) =& Pr(X = i) \\ =& Pr(X = i|Z = 1)Pr(Z = 1) + Pr(X = i|Z = 0)Pr(Z = 0) \\ =& \frac{1}{2}P(i) + \frac{1}{2}Q(i) \\ =& R(i). \end{aligned}$$

Thus, $p_X = R$.

(b) Correct Answer is $I(X;Z) = JS(P||Q)$.
Let $M = \frac{P+Q}{2}$. Note that X is distributed according to M by definition.

$$I(X;Z) = H(X) - H(X|Z) \tag{1}$$
$$= \sum -M_i \log M_i - \frac{1}{2}\left(\sum -P_i \log P_i + \sum -Q_i \log Q_i\right) \tag{2}$$
$$= \sum -\frac{P_i}{2} \log M_i + \sum -\frac{Q_i}{2} \log M_i - \frac{1}{2}\left(\sum -P_i \log P_i + \sum -Q_i \log Q_i\right) \tag{3}$$
$$= \sum \frac{P_i}{2}(\log P_i - \log M_i) + \sum -\frac{Q_i}{2}(\log Q_i - \log M_i) \tag{4}$$
$$= \frac{1}{2}\sum P_i \log \frac{P_i}{M_i} + \frac{1}{2}\sum Q_i \log \frac{Q_i}{M_i} \tag{5}$$
$$= \frac{1}{2}D(P||M) + \frac{1}{2}D(Q||M) \tag{6}$$
$$= JS(P||Q) \tag{7}$$

An alternate proof, which exploits the representation of mutual information in terms of relative entropy is as follows:

$$\begin{aligned} I(X;Z) \overset{(a)}{=}& D(P_{X,Z}||P_X P_Z) \\ =& D(P_{X|Z}||P_X) \\ =& E\big[\log \frac{P_{X|Z}}{P_X}\big] \\ \overset{(b)}{=}& E\left[E\big[\log \frac{P_{X|Z=z}}{P_X}|Z = z\big]\right] \\ \overset{(c)}{=}& \frac{1}{2}D(P||M) + \frac{1}{2}D(Q||M) \\ =& JS(P||Q). \end{aligned}$$

In the above, (a) follows by definition of mutual information. (b) follows by tower property of expectation. (c) follows from the fact that $P_{X|Z=1} = P$ and $P_{X|Z=0} = Q$, and $P_Z(0) = P_Z(1) = \frac{1}{2}$.

(c) Correct answer is 1.
Recall from part 1, that

$$JS(P||Q) = I(X;Z) = H(Z) - H(Z|X) \leq H(Z) = 1.$$